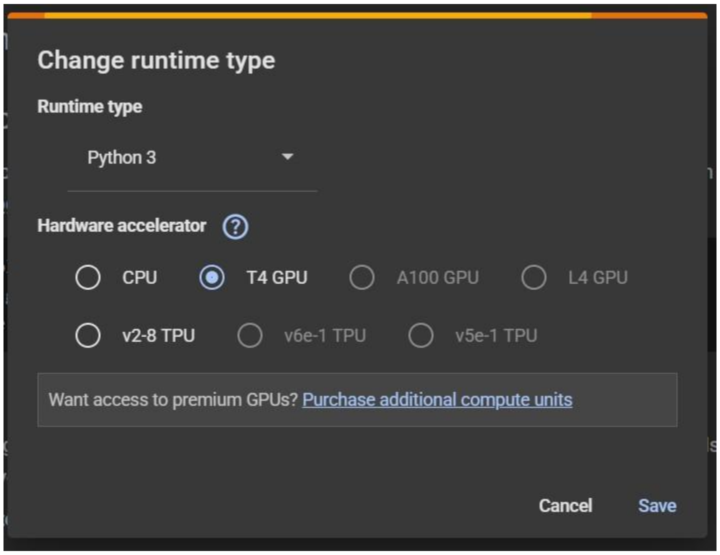
AI/ML -Lab 02

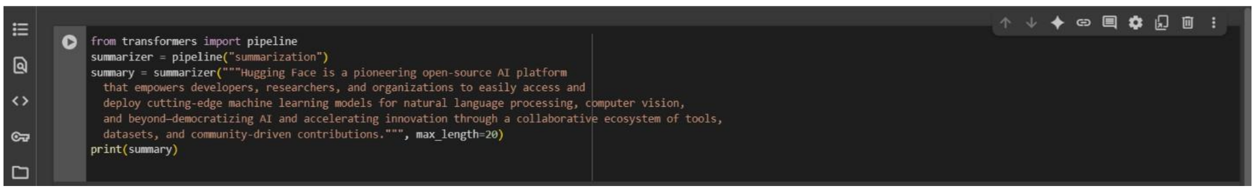
IT24102555-Weerathunga B.A

Exercise 05

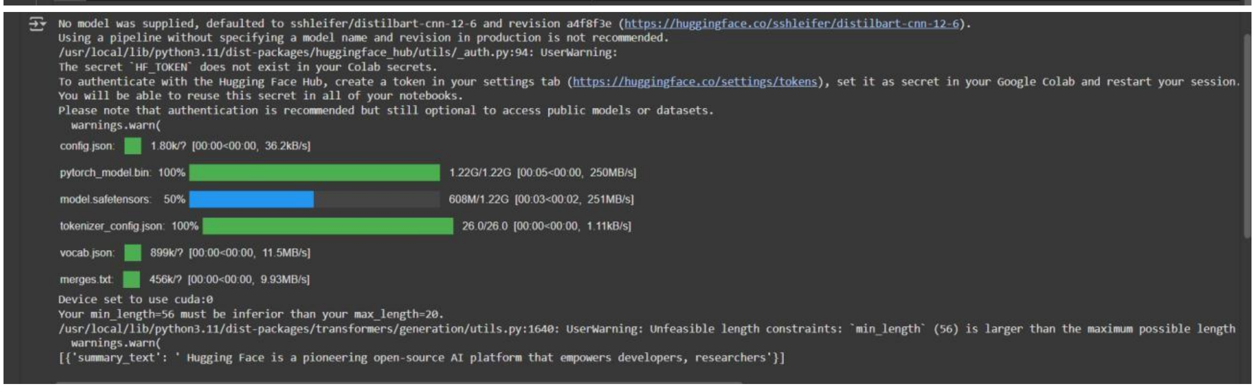

MetaStoneTec/XBai-o4

**XBai o4** is trained based on our proposed **reflective generative form, which combines "Long-CoT Reinforcement Learning" and "Process Reward Learning" into a unified training form**. This form enables a single model to simultaneously achieve deep reasoning and high-quality reasoning trajectory selection. By sharing the backbone network between the PRMs and policy models, XBai o4 significantly reduces the inference cost of PRMs by 99%, resulting in faster and higher-quality responses.



```python
from transformers import pipeline
summarizer = pipeline("summarization")
summary = summarizer("""Hugging Face is a pioneering open-source AI platform
  that empowers developers, researchers, and organizations to easily access and
  deploy cutting-edge machine learning models for natural language processing, computer vision,
  and beyond—democratizing AI and accelerating innovation through a collaborative ecosystem of tools,
  datasets, and community-driven contributions.""", max_length=20)
print(summary)
```



```
No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (https://huggingface.co/sshleifer/distilbart-cnn-12-6).
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 1.80k/? [00:00<00:00, 36.2kB/s]
pytorch_model.bin: 100% 1.22G/1.22G [00:05<00:00, 250MB/s]
model.safetensors: 50% 608M/1.22G [00:03<00:02, 251MB/s]
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 1.11kB/s]
vocab.json: 899k/? [00:00<00:00, 11.5MB/s]
merges.txt: 456k/? [00:00<00:00, 9.93MB/s]
Device set to use cuda:0
Your min_length=56 must be inferior than your max_length=20.
/usr/local/lib/python3.11/dist-packages/transformers/generation/utils.py:1640: UserWarning: Unfeasible length constraints: `min_length` (56) is larger than the maximum possible length
  warnings.warn(
[{'summary_text': ' Hugging Face is a pioneering open-source AI platform that empowers developers, researchers'}]
```

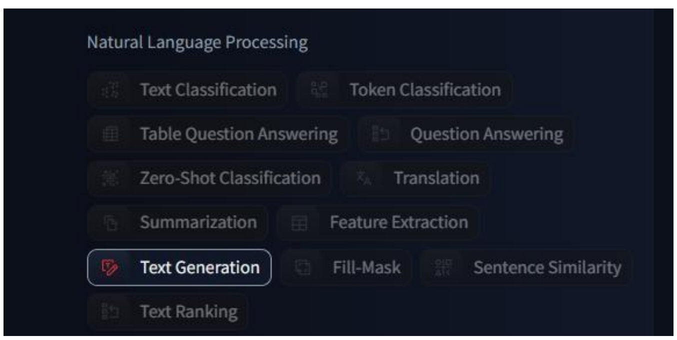
```python
from transformers import pipeline

# Use a known summarization model
summarizer = pipeline("summarization", model="EbanLee/kobart-summary-v3")

# Input text to summarize
text = """Hugging Face is a pioneering open-source AI platform
that empowers developers, researchers, and organizations to easily access and
deploy cutting-edge machine learning models for natural language processing, computer vision,
and beyond—democratizing AI and accelerating innovation through a collaborative ecosystem of tools,
datasets, and community-driven contributions."""

# Generate summary
summary = summarizer(text, max_length=20, min_length=5, do_sample=False)
print(summary)
```

```
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 1.68k/? [00:00<00:00, 99.1kB/s]
You passed `num_labels=3` which is incompatible to the `id2label` map of length `2`.
You passed `num_labels=3` which is incompatible to the `id2label` map of length `2`.
model.safetensors: 100% 496M/496M [00:03<00:00, 236MB/s]
generation_config.json: 100% 191/191 [00:00<00:00, 20.8kB/s]
tokenizer_config.json: 39.5k/? [00:00<00:00, 2.92MB/s]
tokenizer.json: 1.05M/? [00:00<00:00, 36.6MB/s]
special_tokens_map.json: 100% 692/692 [00:00<00:00, 46.6kB/s]
Device set to use cuda:0
Both `max_new_tokens` (=256) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggin
[{'summary_text': 'Hugging Face is a pioneering open-source AI platform\nthat empowers developers, researchers, and organizations to easily access anddeploy cutting-edge machine learni
```




```python
from transformers import pipeline

generator= pipeline("text-generation", model="gpt2")
prompt= "simple java code"
results= generator(prompt, max_length=50, num_return_sequences=1)
print(results[0]['generated_text'])
```

```
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100% 665/665 [00:00<00:00, 28.5kB/s]
model.safetensors: 100% 548M/548M [00:14<00:00, 40.3MB/s]
generation_config.json: 100% 124/124 [00:00<00:00, 7.19kB/s]
tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 1.80kB/s]
vocab.json: 100% 1.04M/1.04M [00:00<00:00, 4.65MB/s]
merges.txt: 100% 456k/456k [00:00<00:00, 1.50MB/s]
tokenizer.json: 100% 1.36M/1.36M [00:00<00:00, 8.90MB/s]
Device set to use cuda:0
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first' truncation strategy. If you encode pairs of sequences (GLUE-style) with the tokenizer you
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)
simple java code to run with the java command.

$ java -jar my.java -jar java.jar -jar my.joutile

Run java from a command line

$ java -jar my.java -jar java.jar -jar java.jar

If you have java installed and run the following command after starting java, it would compile and run the following commands.

$ java -jar my.joutile \ -jar java -jar my.joutile \ -jar java -jar java -jar java -jar

To run the same command with both the java command and the java command you can use the java -jar command.

$ java -jar my.joutile \ -jar java -jar my.joutile \ -jar java -jar java -jar java -jar java -jar

The command will run java with all the following arguments:

java -jar my.jar -jar java -jar my.joutile

This command will run my java command with the following arguments:

java -jar my.java -jar java -jar my.joutile -java -jar java -jar my.jar -java -jar java -jar
```